

Class activity: maximum likelihood estimation

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

Maximum likelihood estimation

Suppose we have a variable Y that can take **three** possible outcomes: $Y = -1, 0$, or 1 . We also know that

- $P(Y_i = 0) = \pi_0$
- $P(Y_i = -1) = 2\pi_0$
- $P(Y_i = 1) = 1 - 3\pi_0$

where the parameter π_0 is unknown. We observe data $-1, -1, 0, 1, 0, -1$, and we want to estimate π_0 from the data.

Questions

1. Write down the likelihood $L(\pi_0)$ for the observed data $-1, -1, 0, 1, 0, -1$.
2. Use calculus to maximize the likelihood from Exercise 1. Remember to differentiate the log likelihood, and recall the following rules for logs and derivatives:
 - $\log(xy) = \log(x) + \log(y)$
 - $\log(x^y) = y \log(x)$
 - $\frac{d}{dx} \log x = \frac{1}{x}$
 - $\frac{d}{dx} c f(x) = c \frac{d}{dx} f(x)$ for constant c

- $\frac{d}{dx}(f(x) + c) = \frac{d}{dx}f(x)$ for constant c
- $\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$